



Projectile Fragmentation of Heavy Ions

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Outline

1. Where does Projectile Fragmentation come from?
2. What is Projectile Fragmentation?
3. Why we need to use Projectile Fragmentation?
4. The practical use of Projectile Fragmentation

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1. Where does Projectile Fragmentation come from?

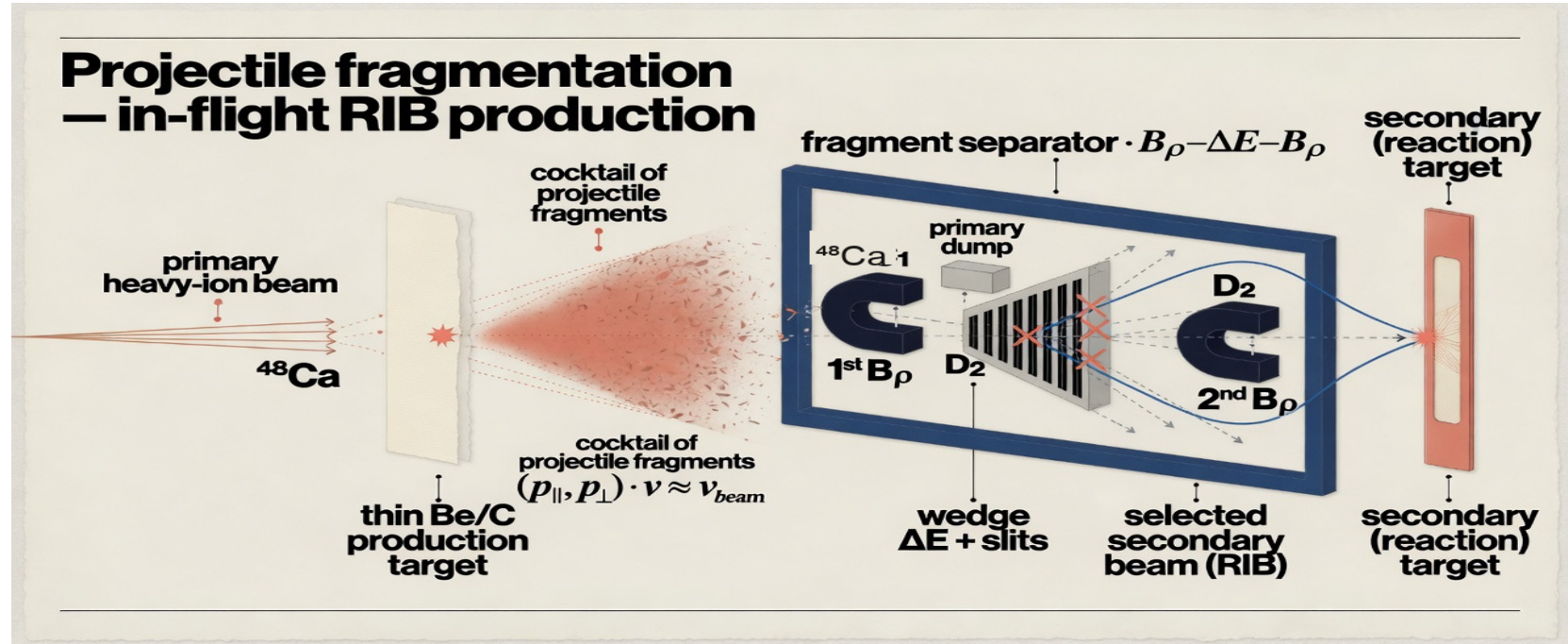
2. What is Projectile Fragmentation?

3. Why we need to use Projectile Fragmentation?

4. The practical use of Projectile Fragmentation

1. Where does Projectile Fragmentation come from?

In principle, any nuclear reaction that produces radioactive nuclei can feed a secondary beam. **It only became practical** for reaction studies after the discovery of **projectile fragmentation** in high-energy heavy-ion collisions.



1. The secondary-beam production chain



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2. What is Projectile Fragmentation?

At beam energy $\gg$ nucleon binding energy, nuclear fragments are emitted into a narrow forward cone — projectile fragments.

1

Many exotic species

Large cross sections for nuclei far from stability — including extreme n-rich and p-rich nuclides first discovered via PF.

2

Nearly beam velocity

All fragments leave with $v \approx v_{\text{beam}}$. Narrow forward cone $\rightarrow$ high collection efficiency and simple magnetic separation.

3

Energy-independent σ

Fragmentation cross section is almost independent of incident energy (once above the relevant scale).

2. Momentum distribution of fragments

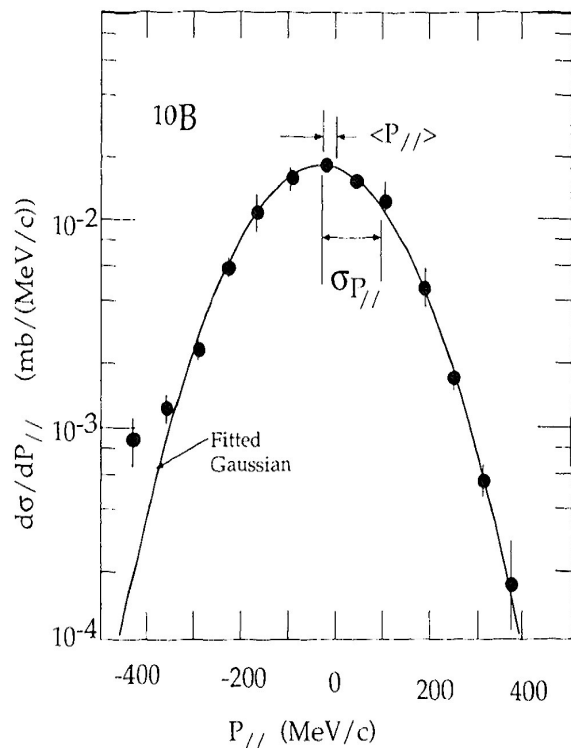


Fig. 1 · ^{10}Be from ^{12}C @ 2.1A GeV (projectile rest frame)

Gaussian shape

Fits all isotopes — independent of beam, energy, or target.

$\sigma_{||} \approx \sigma_{\perp}$

Parallel and transverse widths nearly equal (isotropic in the moving frame).

$\langle P_{||} \rangle$ shift

Central momentum -10 to -130 MeV/c (fragment slightly slower than projectile).

Same velocity

Total fragment momentum $\gg \langle P_{||} \rangle \Rightarrow v_F \approx v_{\text{beam}}$ — the key for separators.

2. Goldhaber width — how broad is the cloud?

$\sigma(P||)$ is essentially independent of target mass and beam energy; it depends on projectile and fragment mass numbers.

$$\sigma(P||) = \sigma_0 * \text{sqrt}[A_F (A_p - A_F) / (A_p - 1)]$$

$\sigma_0 \approx 90 \text{ MeV/c}$ · maximum when $A_F = A_p / 2$

High energy ($\geq 200 \text{ MeV/u}$)

$\sigma(P_t) \approx \sigma(P||)$
Isotropic production in the projectile frame moving at $\beta ||$
 $= -\langle P || \rangle / E$.

Low energy ($< 200 \text{ MeV/u}$)

Extra transverse term with $\sigma_1 \approx 200 \text{ MeV/c}$ (Eq. 2). $\sigma ||$ stays ~constant down to $\sim 20 \text{ MeV/u}$.

2. Why high energy helps collection

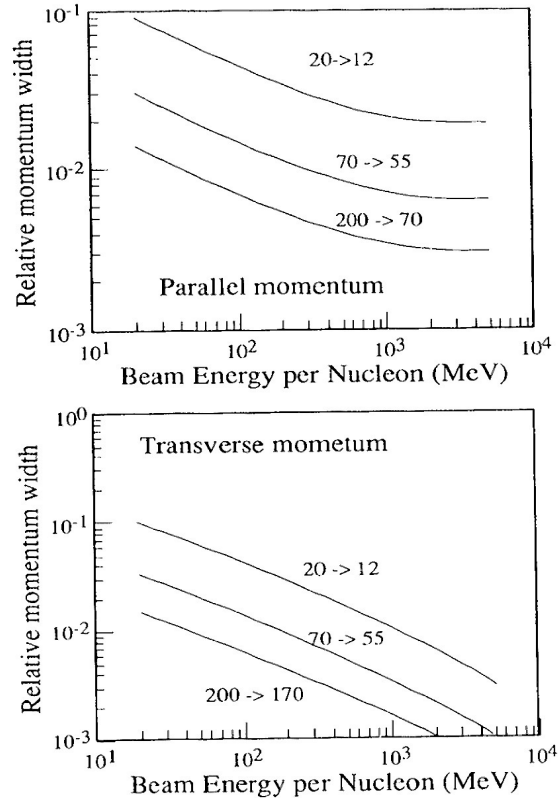


Fig. 2 · Relative momentum width vs beam energy

Relative width shrinks

σ is nearly fixed in MeV/c; total P grows with energy $\rightarrow \sigma/P$ drops.

Narrower lab angles

$\theta \approx p_t / p_{||} \rightarrow$ high energy packs more yield into the separator's $\Delta\Omega$.

Higher collection ϵ

$l_{\text{sec}} = R_{\text{prod}} \times \epsilon_{\text{coll}} \times \dots$ Fixed machine acceptance captures a larger fraction.

Design tip

Choose A_p close to A_F (and $Z_p, N_p \geq Z_F, N_F$) to minimize broadening.

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3. Why we need Projectile Fragmentation

Classical constraint

Stable beam + stable target

Only the narrow valley of stability is well measured:

- radii & moments
- single-particle structure
- giant resonances
- scattering & reaction mechanisms

Vast regions of bound nuclei remain out of reach.

With radioactive beams

β -unstable nucleus in entrance channel

- Structure far from stability (neutron halo, thick skin)
- Exclusive reactions & isospin
- Hot CNO & Big Bang paths
- New isotopes, masses, moments

*PF makes energetic RIB practical for reaction studies
— not only decay spectroscopy.*

3. Same velocity → magnetic separation works

Because fragment velocities are nearly equal, magnetic rigidity maps cleanly onto A/q .

$$B\rho \sim p/q \quad \text{with} \quad p \approx A \cdot m_u \cdot v$$

$$\text{same } \beta \Rightarrow B\rho \sim A/q \quad (\text{fully stripped: } B\rho \sim A/Z)$$

Example: ^{11}Be vs ^{11}C

Fully stripped, same β :

$$A/Z(\text{Be}) = 11/4, \quad A/Z(\text{C}) = 11/6$$

$$(B\rho)_{\text{Be}} / (B\rho)_{\text{C}} = 3/2$$

If velocities differed wildly...

$p = mv$ would scramble. Different (A,Z) would no longer land at distinct $B\rho$ values — magnetic selection would fail.

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4. Practical use — separation principles & target

Three design requirements for efficient secondary-beam delivery

1

Narrow production

Target & reaction chosen so σ_p and σ_θ stay small

2

Large acceptance

Separator Δp and $\Delta\Omega$ as large as practical

3

Achromatic delivery

Momentum spread $\neq$ position smear at physics focus

Target material: prefer low-Z (e.g. Be)

- Lower $dE/dx \rightarrow$ thicker target for same energy loss
- Yield $\sim A^{(-1/3)}$ for fixed g/cm^2 ($\sigma_{frag} \sim A^{(2/3)}$, nuclei/area $\sim 1/A$) $\rightarrow$ light targets win
- Less multiple scattering $\rightarrow$ smaller emittance growth; primary attenuates faster
- Optimum thickness: balance Goldhaber width vs energy-loss Δp ($\geq \sim 5 g/cm^2$, ΔE term rivals reaction width)

4. Separation methods & take-home messages

How species are selected

Bp selection

Same $\beta \Rightarrow R = p/q \sim A/Z$. First stage separates by A/Z at dispersive focus F2.

Energy-loss achromat

Degrader at F2: $dE/dx \sim Z^2 \rightarrow$ second Bp resolves A and Z (not only A/Z).

Wedge shape

$d(x) \sim (1 + x/D_1)^{(2 \cdot \gamma)}$ preserves achromatism for all A, Z, E.

Take-home

1. PF $\rightarrow$ forward, same- v fragments with usable σ far from stability.
2. Cloud width (Goldhaber) sets how much yield fits the separator.
3. Same β makes $Bp \sim A/Z$; degrader adds Z-sensitive purity.
4. Low-Z thick targets + high energy maximize delivered intensity.
5. Facilities: LBL beam line $\rightarrow$ LISE, RIPS, A1900/ARIS, FRS...

Thank you

Questions & discussion

Source: I. Tanihata — Production and Use of Radioactive Beams
(Experimental Techniques in Nuclear Physics, Ch. 10)

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